Sang-Kyun Ko (고상균, 高像均)

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Google scholar | Linkedin | Page | Academic statement

EDUCATION

advanced Application for Intelligence systems Lab.(aAIs Lab), HanYang University (HYU), Seoul, Korea [link]
September 2019 - February 2022; September 2023 - February 2024

- M.S. degree in **Department of Computer Science**
- Thesis: 'Learning to Directly Maximize Evaluation Metrics: Focus on Matching Efficiency and Concordance Index' [link]
- GPA 3.33 / 4.0
- **Teaching Assistant: Artificial Intelligence (CSE4007), Automata Theory (ITE3061), Discrete Mathematics (MAT2020), Creative Software Programming: C++ (ITE1015)**
- 2022, Led research on utilizing censored data as training data in survival analysis
- 2020, Collaborative Research with Particle Physics: Identifying additional b-jets in $t\bar{t}b\bar{b}$ process
- Advisor: Prof. Yung-Kyun Noh

System Software Research Lab.(SSR Lab), Sun Moon University, Asan, Korea
(currently changed to Intelligent System and Software Research Lab., CBNU) [link]
March 2013 - February 2019

- B.S. degree in **Department of Computer Science**
- Graduation Project: 'Deep Learning-based System for Object Retrieval in Multimedia'
- GPA 3.35 / 4.5
- 2017, **Student Excellence Scholarship**
- Advisor: Prof. BongJae Kim

SHORT TERM EDUCATION

- February 2024, Pattern Recognition and Machine Learning Winter School 2024 (12 hours, Online) [link]
- January 2024, 3D Computer Vision and Neural Fields by Prof. Kwang Moo Yi, (9 hours, Online) [link]
- February 2021, Pattern Recognition and Machine Learning Winter School 2021 (8 hours, Online) [link]
- December 2020, The 11th KIAS CAC Online Winter School: The Parallel Computing and Artificial Intelligence (20 hours, Online) [link]
- April 2019, Launching into Machine Learning: an online non-credit course authorized by Google Cloud and offered through Coursera (14 hours, Online) [link]

EMPLOYMENT

Machine Learning Researcher
(Contract with Professor)
Oct 2023 - Ongoing
Institution: Seoul National University Bundang Hospital, SNUBH
Department: Coronary Artery AI team, Cardiovascular Center, Internal Medicine

- Fluoro to Cine: CAG image restoration via rectified flow (Ongoing)
- Automated coronary artery tracking in CAG sequences (Ongoing)
- Video frame interpolation via flow matching (Under review, Co-author)
- Estimating the Risk of Atherosclerotic Cardiovascular Disease Using Health Screening Data (Under review, First author)
- 3D Reconstruction of Coronary Arteries from CAG

Machine Learning Research Engineer
(Full-time employee)
April 2022 - July 2023
Institution: AINEX
Department: Gastroscopic AI Part, Al Research and Development Team

- Endoscopic Image Generation via Diffusion Model
- Gastroscopic Image Analysis for Diagnosis
- Real-time Image classification
- Detected Object Size Estimation in Monocular Lens

Assistant Researcher
(Contract employee)
April 2019 - July 2019
Institution: Industry-University Cooperation Foundation at Sun Moon University
Department: Sytem Software Research Lab.

- Chinese Cursive Character Recognition based on Convolutional Neural Network

CONFERENCES

1. **Sang-Kyun Ko** and Yung-Kyun Noh, (2023), Data Reconstruction Method for Learning to Censored Survival Data, *Korean Artificial Intelligence Association Summer Conference, 2023*, Poster presentation. [Program book link], [Paper], [Poster]
2. Oh, M. J., Park, J., Jeon, J., **Ko, S. K.**, Jo, S., Kang, S., Kim, S. H., Park, S. H., Chang, Y. H., Shin, C. M., Kang, S. J., Kim, S. G., & Cho, S. J. (2023). Application of artificial intelligence in the detection of Borrmann type 4 advanced gastric cancer in upper endoscopy. *United European Gastroenterology Journal, 11(Supplement 8), pp0253, p666.*, Abstract Poster, [Abstract]
3. **Ko, S**., Kim, B., and Kim, J. D., (2018), Deep Learning based Algorithm for Object Identification in Multimedia, *The 13th KIPS International Conference on Ubiquitous Information Technologies and Applications (CUTE 2018)*, Oral Presentation. [Program book link]
4. Park, K., Hong, B., **Ko, S.**, and Kim, B. (2018). Design and Implementation of Real-Time Web Authoring Tool Based on Drag-and-Drop Method. *The 13th KIPS International Conference on Ubiquitous Information Technologies and Applications (CUTE 2018)*, Oral Presentation. [Program book link]

5. Hyeontae Seo, Boseon Hong, **Sang-Kyun Ko**, Kichoel Park, Bongjae Kim, and Hyedong Jung (2017). A survey on the performance of Deep Learning based on multiple GPUs, Poster presentation *Korea Software Congress 2017, Busan.*, [Program book link] [Paper].

PUBLICATIONS

1. (Under review) Kwon, H.*, Park, S.*, Kim, D.H., **Ko, S.K.**, Kim, S.H., Moon, I. T., Jeon, K.H., Kang, S.H., Yoon, C.H., Youn, T.J., Chae, I.H. (2026). Angio-FILM: Video Frame Interpolation for Coronary Angiography using Latent Flow Matching.
2. (Under review, Preprint) **Sang-Kyun, K.**, Si-Hyuck, K., Sun-Hwa, K., Hwi, K., Chang-Hwan, Y., Tae-Jin, Y., & In-Ho, C. (2025). Kolmogorov-Arnold Network for Atherosclerotic Cardiovascular Disease Risk Prediction. medRxiv, 2025-05. [Project page]
3. Moon, I.T.*, **Ko, SK.***, Kang, SH. et al. (2024). Augmented Reality in Cardiology: Enhancing Visualization and Precision. *Current Cardiovascular Risk Reports.* (***contributed equally to this work**.)
4. Jang, C., **Ko, S.**, Choi, J., Lim, J., Noh, Y. K., and Kim, T. J. (2022). Learning to increase matching efficiency in identifying additional b-jets in the $t\bar{t}b\bar{b}$ process.*The European Physical Journal Plus,137(7), 1-12.*

AWARDS

- Nov 2018, **Excellence prize**, 4th Industrial Revolution Talent Development Joint Festival in the fields of Papers or Coding, awarded by *Korea Society of Information Communication Security Ethics*
- Nov 2018, **Grand Prize**, Comprehensive Competition for Union of Computer Science and Electrical Engineering, awarded by the *Department of Computer Science at Sun Moon University*
- June 2018, **Silver Prize**, Comprehensive Competition for Computer Science, awarded by the *Department of Computer Science at Sun Moon University*
- Nov 2017, **Bronze Prize**, Global capston design workshop, awarded by *Idea Factory at Sun Moon University*
- April 2017, **Excellence prize**, Idea Contest held by the President, awarded by *President of Sun Moon University*

CERTIFICATES

- May 2019, Engineer Information Processing(정보처리기사), Human Resources Development Service of Korea
- June 2016, Driver's license, Gyeonggi Nambu Provincial Police Agency

RESEARCH INTERESTS

- Addressing theoretical and statistical issues in machine learning for large scale data analysis
- Interested in flow-based generative modeling, with a focus on robustness under distribution shift
- Being interested in diverse topics, including multi-modality, sequence models, and large-scale generative models

MILITARY SERVICE

March 2014 - December 2015
Infantry of Army, Yanggu-gun, Gangwon-do, Republic of Korea

- PRC-999K Platoon Signalman and K3 Assistant Gunner

References

- **Yung-Kyun Noh** (nohyung@hanyang.ac.kr)
  Professor, Computer Science, Hanyang University

- **BongJae Kim** (bjkim@chungbuk.ac.kr)
  Professor, Computer Science, Chungbuk National University